Underwater Vision Profiler (UVP5) Standard Operation Protocol for Bio-GO-SHIP

Version 1.0

Authors

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Protocol Revision Record

Version	Date Implemented	Description of Revisions
1.0	3 December 2024	Initial Release

Acronyms, Abbreviations, Terminology

Term	Definition
GO-SHIP	Global Ocean Ship-Based Hydrographic Investigations Program
Bio-GO-SHIP	Biological GO-SHIP
UVP/UVP5	Underwater Vision Profiler 5
CTD Niskin rosette	Conductivity, temperature, depth (CTD) sensor attached to a metal rosette frame for Niskin bottles

Background

Summary

This protocol describes considerations for the use of the UVP5 during pre-mobilization, at-sea protocol, and post-mobilization of Bio-GO-SHIP research cruises.

Spatial coverage and environment(s) of relevance

This protocol can be used across any marine environment down to 6000 m to effectively collect continuous UVP data from a CTD Niskin rosette.

Ship space requirements

The UVP will be mounted to the CTD rosette. The UVP is internally powered, and data is internally logged. The power and data cables will be fed from the CTD hanger to the Wet Lab. In the Wet Lab, the UVP deck unit and laptop will be secured on available counter space. There is an analog cable from the UVP to the CTD that allows for the UVP voltage to be read from the computer room live during the cast. This voltage is not recorded and can be thought of as a proxy for particle abundance but is not calibrated for any scientific purpose other than determining that the UVP appears to be operating as expected, or not, during casts.

Personnel Required

One person onboard is trained as the UVP Trained Operator. One additional person is trained in preparing the UVP for deployment, recovery, and charging; this person can be trained during mobilization. These two people should work opposite shifts to ensure one person is present for each CTD cast.

Safety

The lithium-ion battery is considered HAZMAT. HAZMAT code: UN 3481 Lithium-Ion Batteries Contained in Equipment. The battery weighs 1.09 kg.

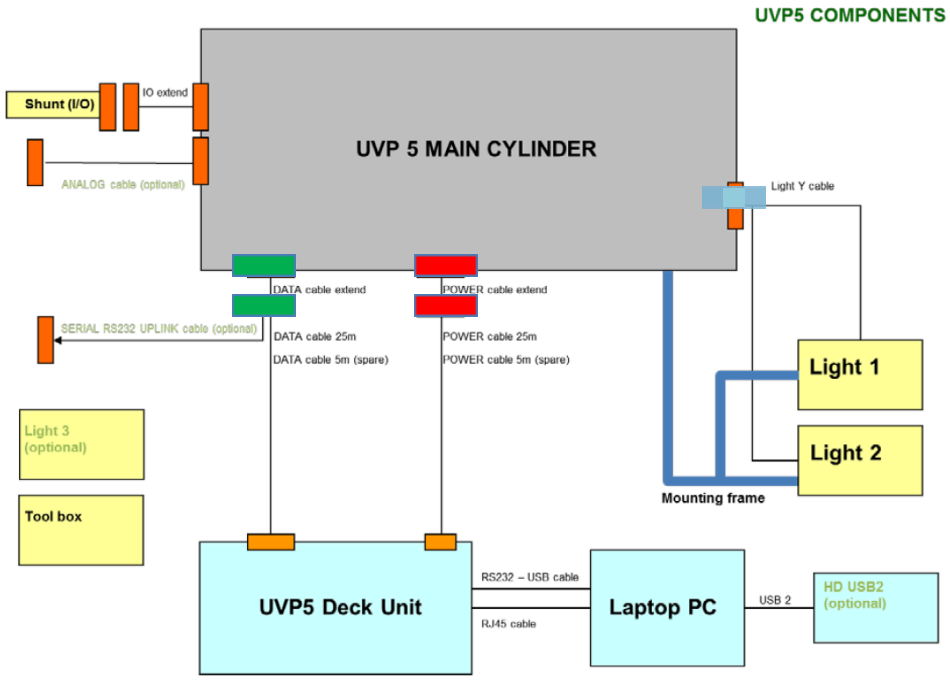
Take caution when lifting and moving the UVP main housing and lighting unit. At least two people capable of lifting 50 lbs should assist in assembling, mounting, and disassembling the UVP.

The UVP should be rinsed and quickly inspected after each cast to ensure that there are no cracks in the lighting unit, no flooding of the instrument, and no corrosion or water intrusion of any cables. If these occur, they can pose safety risks such as contents under high pressure or fire hazards.

Pre-mobilization

The UVP instrument, cables, tools, and laptop are stored in a single Pelican case. A printed checklist of all contents should be included.

Instrument specifications



The UVP consists of the main camera/pressure housing unit and lighting unit, which connects to a deck unit via power (red) and data/sea (green) cables. The main camera bulkhead has 1 m power and data cable extensions that attach to 25 m power and data cables. The extensions stay attached to the UVP during the cruise to avoid excess wear to the instrument ports. The analog cable can connect the UVP to the CTD to provide displays of the UVP voltage on ship computers.

Shipping the instrument

It is not recommended to remove the lithium-ion battery from the UVP. Shipping with the battery is classified as HAZMAT.

Power: 29.4 Vdc / 6.8 A / 199.92 Wh
HAZMAT code: UN 3481
Lithium-Ion Batteries Contained in Equipment 1.09 kg

If shipping from UCI, proper loan agreement documents must be signed by both Adam Martiny and Emmanuel Boss. Please contact Equipment Manager (Helen Chang,

hmchang@uci.edu). The remaining shipping details are coordinated with Purchasing Manager (Jeff Ledendecker, jledende@uci.edu).

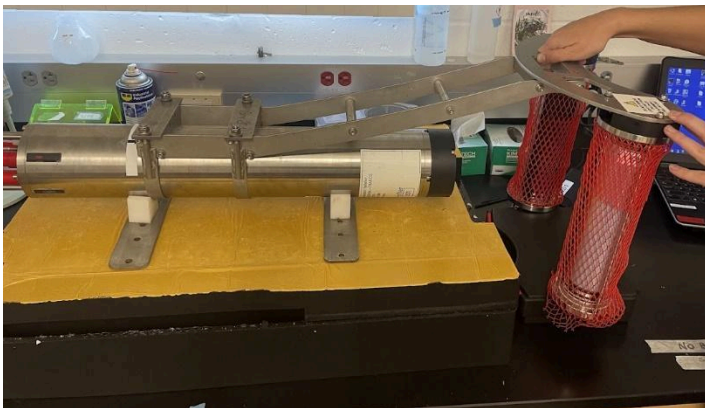
Bio-GO-SHIP Shared Drive Links: Shipping Recommendations, Annotated UVP5 User Manual

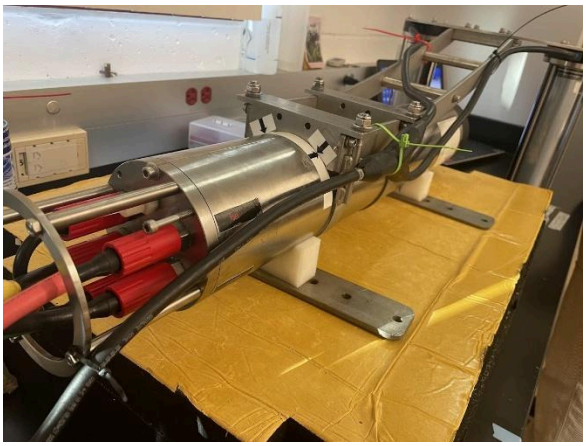
Remove grease, pressure oil bottle, and the laptop from the UVP Pelican case before shipping and pack these items with personal bags. These items have causes issues for us in the past.

At sea

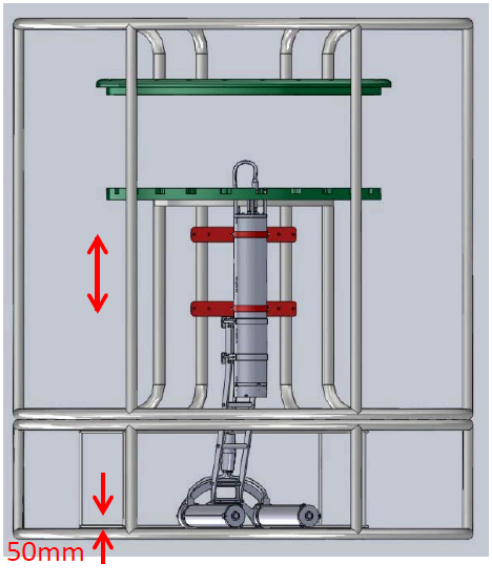
Assembling and mounting UVP

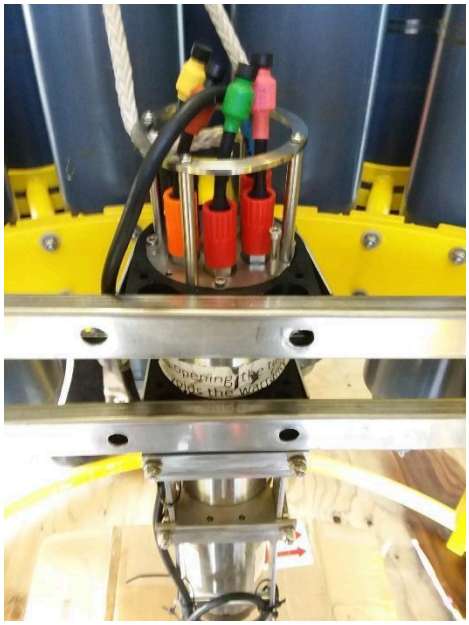
Have at least two people capable of lifting 50 lbs when assembling the UVP.





Following manual page 16, assemble UVP on a countertop, using the foam packing level and CTD mounting cradles to stabilize instrument. Toolbox may be used to support lighting unit. Secure lighting cable with zip ties as shown. Remove tools from the toolbox that you may need to use during assembly. Remove dummy connections from the UVP bulkhead, wipe away any grease from the metal pins, and apply thin layer of new grease to the female adapter ends. Use kim wipes to remove any gunk and build up around pins if needed. Ensure that adapters are tightly seated before tightening red caps.







Work with CTD technicians to mount UVP with their own mounting hardware, or with mounting hardware included in UVP Pelican case. When mounting the UVP to the rosette, be very careful to ensure nothing is damaged.

- The glass lights are very fragile, so cover them with a piece of foam or a towel when moving the UVP.
- Place foam or anti-skid mats on the ground below where you are working in case there is contact.
- Use the overhead winch with a line on the UVP bulkhead cage to help position the UVP and take weight.
- After the UVP has been mounted, transfer this line as a secondary security, secure UVP bulkhead cage to top of rosette frame with a line in case the mounting hardware fails.
- Check distance between mounting bars and rosette frame regularly to ensure no slippage. If you notice that the instrument is slipping, tighten all mounting bolts or remount if needed. If slippage occurs unnoticed, then the lighting units may hang too low and be damaged during CTD recovery process.
- Ensure that the lighting units are secured 4-5 cm above the bottom bar of the rosette frame but also have no obstructions below the lighting units. This ensures water shear is minimized as water passes into the sampling area between the lights.

After the UVP has been mounted, double check the distance between the UVP pressure case and the lighting units using the T bar and metal ruler. Plus or minus one mm is fine, as it is hanging now, but ensure the lights are securely attached and aren't slipping down the shaft of the pressure housing cylinder.

The deck box and laptop will be secured in the Wet Lab. The 25 m power and data cables will run from the deck box to the CTD hanger, secured in a manner that keeps them out of the way

of other operations. Ensure there is sufficient space in the small port hole to pass these wires through. The port holes can get filled with many cables and the larger ends need to be fed through first.

Check UVP connection

- Do not connect the laptop to internet.
- Connect UVP to deck box with power and data cables. Connect deck box to the laptop with USB and ethernet cables. Ensure to connect the RS232 USB cable to the USB port marked with a red dot on the right side of the laptop.
- Turn on deck box power switch on left side of back panel, the red power switch, and the voltage (VU) switch. On the deckbox switches, 1 is on and 0 is off. Use diagrams on page 26 and 27 to connect the UVP5HD to the Deck unit.
- Check com port number in *view devices and printers*. The instrument will be called EdgePort Serial Port (COMx). Com port must be a single digit number. If it is 10 or above it will need to be reassigned to a different port number.
- Check configuration folders for project in C: drive
- Open ImageJ and the cruise project should be already created and listed in the dropdown menu
    - This project is already configured to acquire data utilizing depth mode with the current UVP calibration files
- Enter UVP Pilot Tools
- If there are errors at this point, then there may be issues with comm port, cable connections, instrument, or deck box.
- Ensure that laptop is appropriately set to UTC time and that the date and time are correct.
    - If not, connect the laptop to a wifi network, go to the control panel, search for set date and time, click Internet time, and synchronize the internet time. I used windows.time.com previously and that fixed the time previously. Check that the date and time match UTC within +/1 1 second. Disconnect laptop from the wifi network and forget network or prevent the laptop from auto-connecting to the wifi network.
    - After ensuring the laptop time is correct, in ImageJ enter UVP Pilot Tools, select operation Synchronize UVP5 time with Computer time
- Perform light beam test as outlined on manual page 80 – 82.
- In UVP Pilot Tools, select operation EXIT PILOT MODULE, follow prompts
- Okay to close ImageJ

Additional setup tasks

- Check lights for damaged bulbs
    - Open ImageJ and enter UVP Pilot Tools
    - Select module <TURN ON UVP5 light (test alignment)

    - As lights are on for 30 seconds, run your hand through each beam to ensure no discontinuity
- Check beam parallelism
    - Follow same steps above to test light alignment
    - Insert Tshape jig between both lights, and beam should be focused in the middle of central line
    - If not aligned properly, follow manual instructions to adjust parallelism

Operation notes

When lowering the CTD to soak ato 22 meters depth, the descent speed needs to be at least 0.3 m/s, ideally 0.5 m/s if possible. If the cast needs to be aborted during the soak, we request that the rosette is lowered to 35 m prior to being brought onboard- this assures the UVP turns off in the pressure operated programming. Make sure UVP is fully charged prior to deployment.

If there is sufficient battery (Vu > 26) the UVP will sample during all GO-SHIP casts in depth mode using the Power Shunt. The UVP will not sample during the Bio-Go-SHIP casts to allow for sufficient battery charge for full casts and to not repeat sampling at the same station. After noting the final voltage, turning off the charger and deck box, and disconnecting the power and/or data cables, then the Power Dummy should be plugged into the 1 m Power Extension Cable and secured rather than the Power Shunt. Without the power shunt, the UVP will not turn on even if the proper soaking sequence is performed.

If any issues occur with the UVP and you do not want to run it on a cast, but you don't have time to remove it from the rosette, you can send it in with the Power Dummy. However, I would encourage you not to do this too many times as it could indicate flooding which wouldn't be safe to send to depth and back.

Deployment

Pre-deployment (every cast)

- Turn off battery charger (green switch) on deck box
    - Leave UVP charging as long as possible, but do not delay CTD operations
    - Note the final voltage (VU) in the excel sheet in column F. It must be more than 26 vu for there to be sufficient battery to run during the cast.
- Turn off voltage (VU) switch on deck box
- If on, turn off external power (red switch) on deck box. This should only be on if data was previously being downloaded or deleted and should be turned off when those processes are complete. There is no need to have external power on for charging the instrument.
- Turn off deck box with switch on left side of deck box back panel
- Disconnect power (red) and data (green) 25 m cables from the 1 m extensions

- Connect red power shunt to power extension, and connect green dummy to data extension. Check that these connections are fully seated before screwing on the plastic connectors.
- Secure extension cables to the inside of the CTD frame with velcro. Check that all cables are tightly secured and not at risk of being pinched in a niskin bottle or being crushed between the rosette frame and the side of the ship if there is an impact event.

Deployment

- Fill in station data sheet
    - Yellow columns should be filled out before the rosette is deployed
        - Cruise (sn201_2025_i09n) and Ship Name (rv_thompson) will be the same for the whole cruise.
        - UVP Operator Name: record the name of whoever is preparing the UVP for deployment.
        - Profile Date (UTC) (YYYYMMDD): example 20250315. This is the date in UTC at the time of deployment of the rosette. This would be March 15, 2025.
        - Approx Profile Start Time (UTC) (HHSS): Record the time you turn off the charger and plug in the power shunt preparing the UVP to be deployed. 1536 = 3:36 PM UTC. Use the time on the UVP Laptop.
        - Battery voltage before cast: after you turn off the charger, record the VU value in the __ window of the deckbox. There should be at least 26 volts for the UVP to run on the next cast. If there aren't 26 volts, plug in the power cable dummy and not the power shunt on the 1 m power cable extension cable.
    - Orange column: G- fill this out after the rosette is recovered after each cast
        - Battery voltage after cast: when you plug in the power cable to the UVP and turn on the deckbox and the VU meter you should be able to read the final voltage of the instrument. Record that value here before you turn on the charger. Then turn on the charger and let the instrument charge. The voltage should be less than it was at the beginning of the cast. It shouldn't go below 23 VU.
    - Green columns can be filled out during or after the cast by referencing the CTD logsheet in the computer lab. May be available on elog if cruise is updating it.
        - Bottom depth (m): the seafloor depth in m that's recorded on the CTD logsheet
        - Latitude (decimal degrees): copy from the CTD log sheet
        - Latitude (DD°): the whole digit latitude as a positive number (should range between 0 and 90)
        - Latitude (MM.MMM): will be auto-calculated from columns I and J.
        - Latitude (N/S): mark if you're in the northern hemisphere (N) or southern (S)
        - Longitude (decimal degrees): copy from the CTD log sheet

        - Longitude (DDD°): the whole digit latitude as a positive number (should range between 0 and 180)
        - Longitude (MM.MMM): will be auto-calculated from columns M and N.
        - Longitude (N/S): mark if you're in the eastern hemisphere (E) or western (W)- I think all of I09n is in the eastern hemisphere
        - Profile ID (i09n_ctdxxxyy) xxx = stn, yy = cast: i09n_ctd will be the same for this whole cruise. Take the three digit station number as xxx (if it's the first station the three digit number would be 001 if it's the 125 station the three digit number would be 125). This should match the CTD logsheet. Then the two digit cast number will likely either be 01 or 02, but it can get higher if the cast is aborted. If the UVP doesn't turn on keep updating the profile number to match the full cast. If the UVP does turn on and off on an aborted cast leave that info and the aborted cast number in a row and create a new row for the new cast on the UVP with the new cast number. Make sure it matched what is on the log sheet.
        - Station ID: the same as xxx from the previous bullet. 001 for station 1 and 125 for station 125. It should match the CTD log sheet. Be careful that leading zeros show up in this column. I have it set as Text to allow for leading zeros.
        - CTD-Rosette reference file should auto fill to equal the Profile ID column.
    - Blue columns: Star will fill these out after you download the data from the instrument and process the metadata.
        - Profile Name: should match the raw folder name for the cast- use the date and time from columns D and E to match up the correct data file to the correct row on this logbook
        - Maximum profile depth (m): this might be recorded on the CTD logsheet. It should be about 5 meters above the seafloor. It's possible we don't need this piece of data, but it's in the example logbook template in the manual which is why I put it here. Use your judgement about if you want to record this or not.
        - First image OK- take from the metadata file firstimage. It should be a whole number ranging from about 2000 – 10000 depending on how long the soak took.
        - Last image OK- take from the metadata file endimg. It should be a whole number that's greater than 10000 and possibly over 100000 if there are a lot of particles in the water or if the cast was very deep.
        - Last Light Test: (YYYY/MM/DD HH:MM) (UTC)- record the approximate time and date of the last time you performed a light test and checked for parallelism. You should do this test once before the first cast then about once per week while you're at sea. This just helps you keep track of when you need to do another test.
- CTD must descend to soaking depth of 22 m at a speed of 0.3 m/s or faster to trigger UVP depth mode setting

- Soak at 22 m for at least 1 minute
- As CTD enters the water, keep an eye on analog voltage reading
    - Prior to soak and during beginning of soak, voltage reading will be lower and toggling between two values
    - During soak, the instrument will turn on and voltage will go up and do large oscillations at higher voltage values
- Re-do soaking if UVP does not turn on (no need to bring back on deck)
- If the cast must be aborted after the soak and the UVP has turned on, the CTD should descend to a depth of 35 m to trigger the UVP to turn off prior to being brought to surface
- Following the soak, CTD comes to the surface, and then begins full descent.
- The UVP will turn off when it recognizes a decrease in pressure of 35 m, usually occurring after the bottom-depth niskin is closed and before reaching the depth of the next niskin bottle on the upcast.

Recovery

- Once CTD is secured on deck, check that UVP is no longer flashing red
- Remove power shunt and reconnect the power cable (red)
- Pack the power shunt in UVP plug bag so it does not get lost
- Rinse UVP with fresh water, especially in small portholes near camera lens and glass lighting units
- Turn on deck box with back switch, voltage (VU) switch and record final voltage reading on datasheet (column G)
- Turn on green charging switch

Troubleshooting

- If UVP comes on deck flashing red, the instrument is still recording
    - **Do not remove power shunt!**
    - Once secured on deck, remove green data cable dummy and attach 25 m data cable
    - Turn on deck box power switch (left side of back panel)
    - Open ImageJ with appropriate project
        - From dropdown menu, select UVP5 Piloting Tools
        - Click OK
        - Prompt appears “Camera is in Acquisition Mode- Would you like to stop acquisition?”
        - Select yes and wait for confirmation of acquisition ending
        - Exit ImageJ

        - Remove red power shunt, reconnect power cable, turn on green charging switch

Once-Daily Tasks

- One of the two Bio techs will be in charge of the following daily and weekly tasks. This person should be trained by a skilled UVP operator and be a skilled UVP operator themselves.
- Check laptop is correctly displaying UTC time
- Near beginning of shift and during transit, back up raw data files to external hard drive and delete files from UVP
    - Connect UVP to green data cable and open ImageJ
    - Select the current project and enter UVP Pilot Tools
    - Select operation DOWNLOAD DATA from UVP5, follow prompts
    - This is saving files in raw folder of project on C drive
    - After downloading is complete, check for photographs .bmp, .dat and .bru files (same number of each), .hdr, and .tsv. If there are more .bmp files, there will be more .dat/.bru files.
    - This takes hours, and you can charge instrument during this time but not as efficiently, so manage time between casts. Do this as close to beginning of shift as possible, and repeat after an additional cast if possible.
    - Back up raw data to external drive (back up the whole project folder, but after downloading data you only need to copy the new cast folders
    - Select operation DELETE DATA IN UVP5, follow prompts
    - In UVP Pilot Tools, select operation EXIT PILOT MODULE, follow prompts
    - Okay to close ImageJ
    - Repeat this, if possible, toward end of shift in order to minimize the amount of time it takes to download and delete data (can take hours to delete data)

Weekly Tasks

- Check lights for damaged bulbs
- Check beam parallelism/light check

Metadata and initial data processing

After data has been downloaded, backed up, and deleted fill out the metadata using the Zooprocess GUI interface. It's easiest to do this daily, but if you have to skip a day and catch up the next day that's ok.

1. Open Zooprocess and enter 2025_i09n project
2. Select fill in metadata from the dropdown menu

3. Cruise: I think this will be auto-filled out based on the project name. It should be sn201_2025_i09n
4. Ship: This might also be auto-filled out: it should be “rv_thompson”
5. Profile ID: i09n_ctdxxxyy where xxx is the three digit station number (example 001 or 025) and yy is the cast number (example 01 or 02). The xxxyy number should match what’s on the CTD logsheet.
6. StationID: the three digit station number (xxx), for example 001 or 025.
7. Bottom Depth (m): take this from the CTD log sheet, example 4749
8. CTD reference (filename): same as Profile ID
9. Latitude (degree): one or two-digit positive whole number from 0 – 90. Take from the Latitude (DD°) column on the UVP Logbook. Don’t put a negative number here for the southern hemisphere.
10. Latitude (minute): Will be auto-calculated in the Latitude (MM.MMM) column. Calculated by taking the latitude in decimal degrees and subtracting the whole number degrees so you just have the decimal points and multiplying this by 60.
    a. Example: (13.975432 – 13)*60 = 58.52592.
11. Latitude (N/S): Mark which hemisphere you’re in either North or South, don’t put negative numbers in the Latitude (degree) column.
12. Longitude (degree): One, two, or three digit positive whole number from 0 – 180. Take from the Longitude (DDD°) column on the UVP Logbook. Don’t put a negative number here for the western hemisphere. I think all of I09n will be in the eastern hemisphere.
13. Longitude (minute): Will be auto-calculated in the Longitude (MM.MMM) column. Calculated by taking the longitude in decimal degrees and subtracting the whole number degrees so you just have the decimal points and multiplying this by 60.
    a. Example: (90.3567866 – 90)*60 = 21.407196.
14. Longitude (E/W): Mark which hemisphere you’re in, either Eastern or Western. Don’t put negative numbers in the Longitude (degree) column.
15. First Image OK: just leave what is auto-selected. If it seems wrong, you can put a note in the comment field at the bottom.
16. Last Image OK: just leave what is auto-selected.
17. Single Descent, yoyo, horizontal, single ascent: select Decent (First Image and auto max depth)
18. Day/Night: no need to select anything here
19. Wind dir (degree): leave whatever is autofilled or leave it blank
20. Wind speed (knts): leave whatever is autofilled or leave it blank
21. Seastate: leave whatever is autofilled or leave it blank
22. Nebuloussness: leave whatever is autofilled or leave it blank

Post-mobilization

Fully rinse the whole UVP and cabling (with dummies connected) to remove all salt water. The 25 m cables probably don’t need to be rinsed. Consider wiping the UVP and 1 m extension cables down with a towel. Wipe the glass lights and the glass camera face with kim wipes. Allow to dry completely before packing in the pelican case.

Carefully remove the UVP from the rosette ensuring the glass lighting units are not damaged. I like to place foam or anti skid mats on the ground below where you are working in case there is contact.

Ideally, the battery should be fully charged, all data downloaded, backed up, and deleted from the instrument before it is packed up. Make sure all of the mounting hardware is returned to its owner (ODF or ours) an dis not left on the rosette. Ensure all nuts, washers, and plastic washers are returned to their homes (on u bolts etc) and tightened finger tight so nothing unscrews or gets lost.

Ensure that all shipping paperwork and HAZMAT considerations are completed for return shipping. Data will be managed by the PI (Stephanie O'Daly).

Remove grease, pressure oil bottle, and the laptop from the UVP Pelican case before shipping and pack these items with personal bags. When traveling, keep the laptop and the hard drive with the data backup in separate bags in case one gets lost or damaged, both copies of data are not lost.

UVP Deployment and Recovery Cheat Sheet

Every Cast

Pre-deployment

- Turn off battery charger (green switch) on deck box, note voltage on elog spreadsheet
- Voltage needs to be 26 or higher to operate for a full cast. Charge of 23 indicates issues with instrument.
- Turn off voltage (VU) switch on deck box
- Turn off external power (red switch) on deck box
- Turn off deck box with switch on left side of deck box back panel
- Disconnect power (red) and data (green, if needed) 25 m cables from the 1 m extensions
- Connect red power shunt to 1 m power extension cable and connect green dummy to data extension. Ensure that adapters are fully seated and watertight, tighten red cap
- For Bio-Casts or other casts where UVP should not be run (i.e. voltage <26), put red dummy on 1 m power extension cable (rather than power shunt)
- Secure extension cables to inside of CTD frame with velcro



For UVP deployment, ensure that the red power cable 1 m extension has Shunt cap secured, and the green data cable 1 m extension has dummy secured.

Deployment

- Fill in station data sheet
- As CTD enters the water, keep an eye on analog voltage reading. CTD watchstanders will mentally note UVP voltage
- Re-do soaking if UVP does not turn on (no need to bring back on deck)
- If the cast must be aborted after the soak, CTD should descend to a depth of 35 m

Recovery

- Check that UVP is no longer flashing red. If still flashing, **do not remove power shunt**, follow troubleshooting protocol
- Visually inspect instrument for slippage and damage
- Rinse UVP (including power shunt) with fresh water, especially in small portholes near camera lens
- Remove power shunt or dummy and reconnect power cable (red)
- Turn on deck box with back switch, voltage (VU) switch and record voltage reading on datasheet
- Turn on green charging switch